

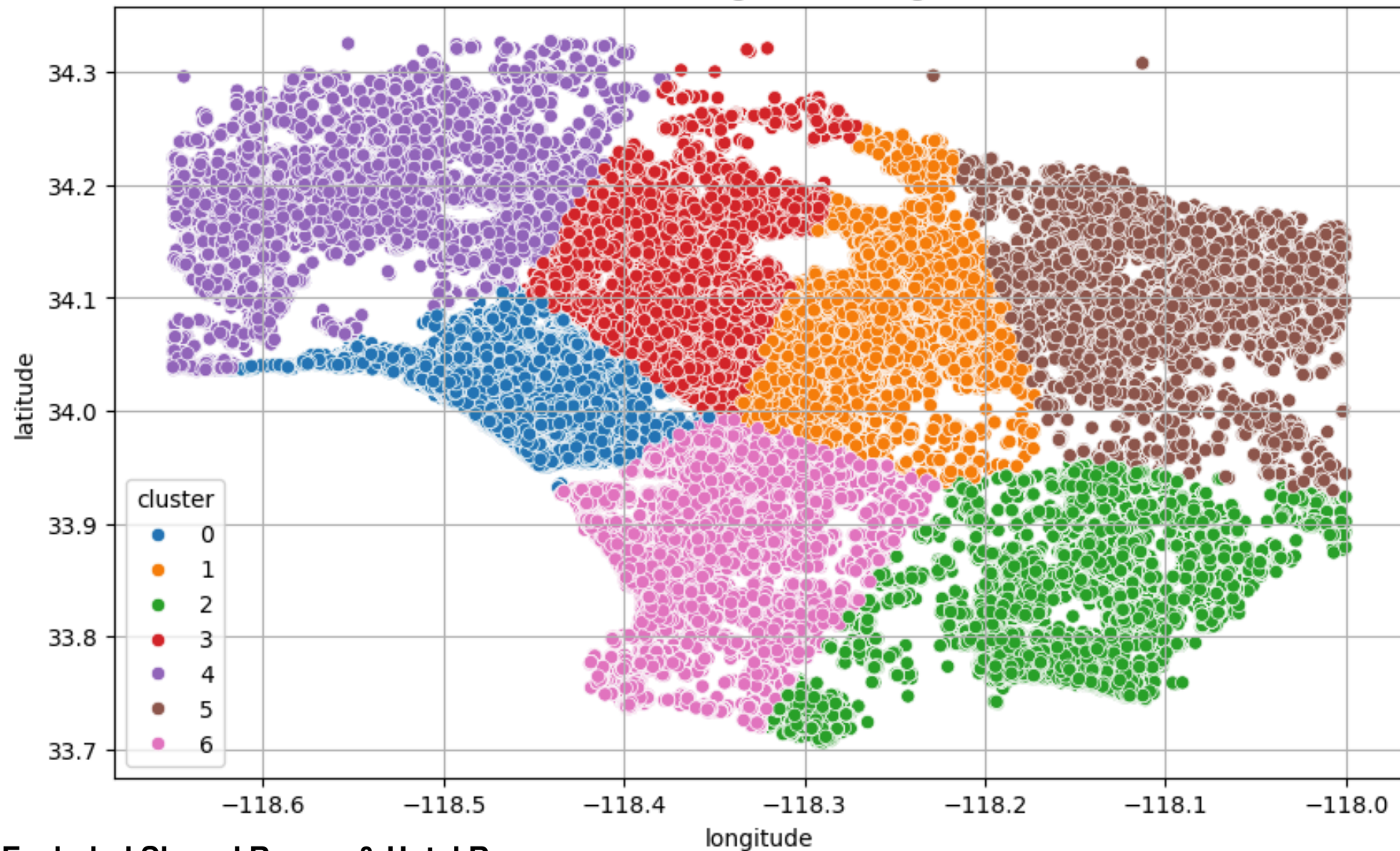


ANALYZING THE SHARING ECONOMY WITH AIRBNB DATA

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Data Selection & Clustering Analysis

KMeans Clustering of LA Listings (k=7)



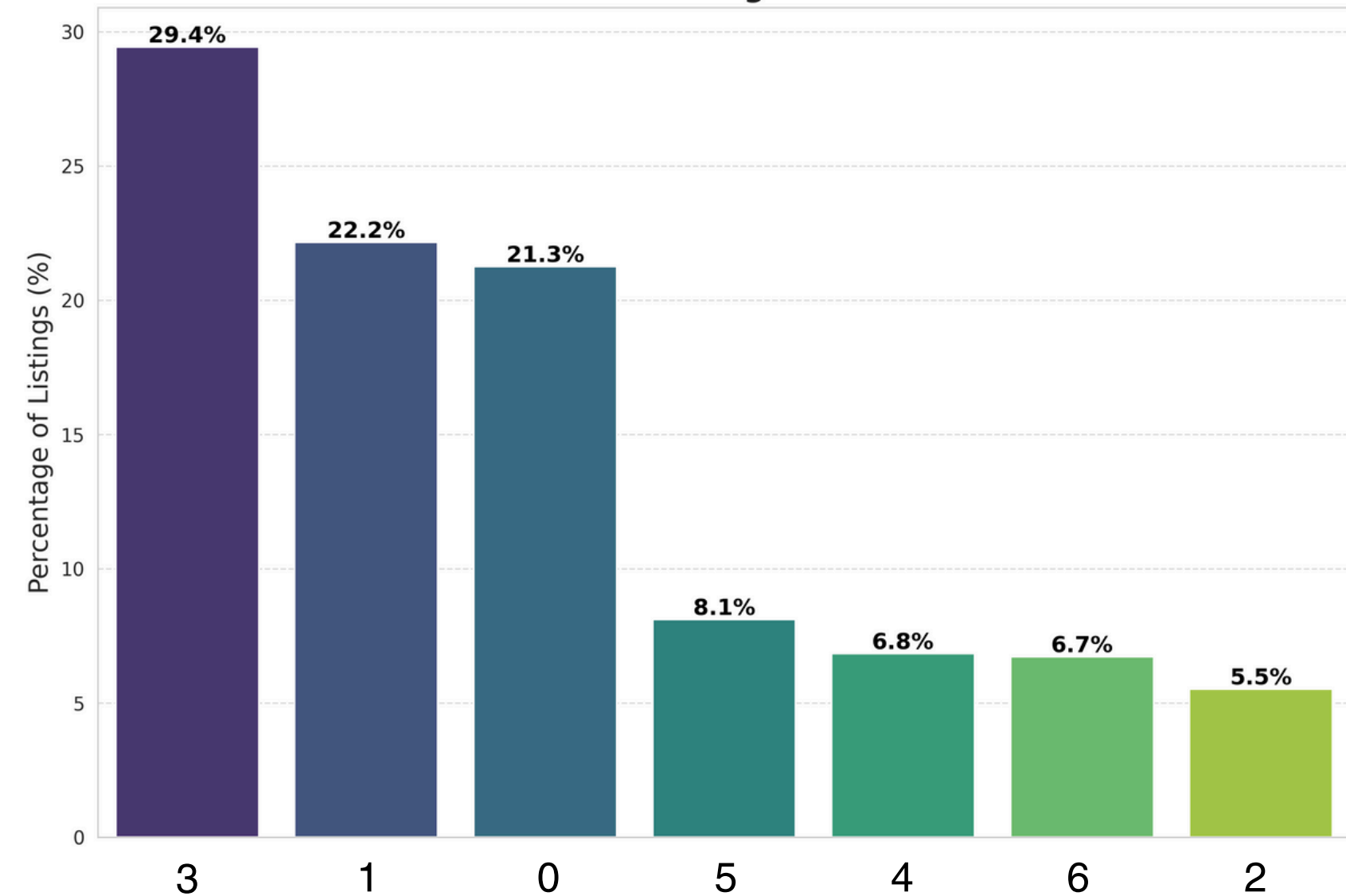
Excluded Shared Rooms & Hotel Rooms

- Focused on private stays: Airbnb primarily caters to short-term renters seeking entire homes or private rooms.
- Avoided commercial listings: Hotel rooms are managed by businesses rather than individual hosts, making them less relevant to peer-to-peer rentals.
- Ensured data consistency: Shared rooms and hotel rooms have different pricing structures and availability patterns, which could skew the analysis.

Focused on the LA region, excluding San Bernardino, Catalina, Malibu, and Palmdale to keep the analysis relevant.

Applied K-Means clustering (k=7) to segment listings into distinct geographic markets.

Distribution of Listings Across Clusters

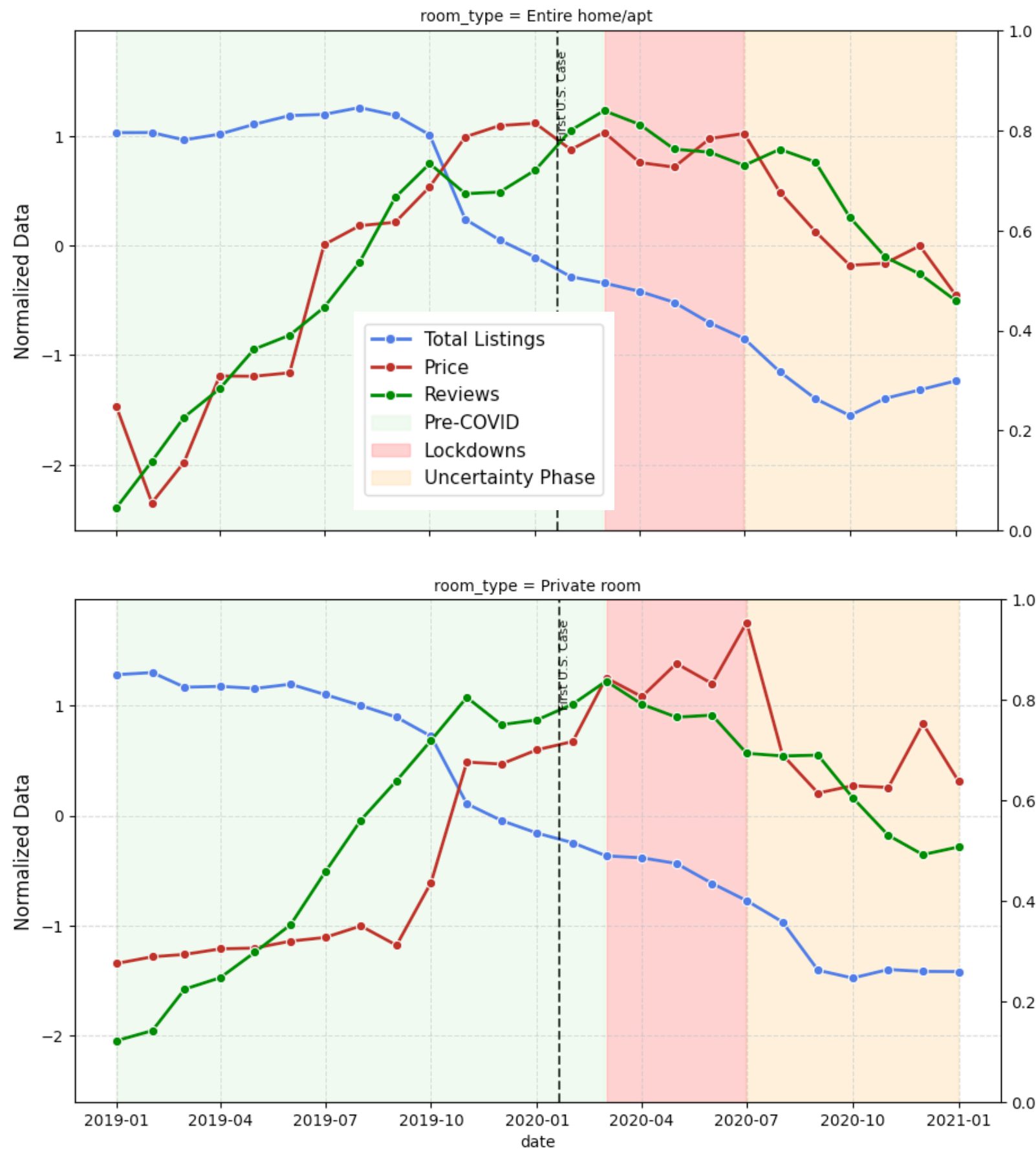


Identified top 3 clusters with the most listings:

- Downtown LA, Hollywood, Glendale (29.4%)
- Pasadena, Altadena, Alhambra (22.2%)
- Santa Monica, Beverly Hills, West LA (21.3%)

These top 3 clusters account for nearly **73% of all Airbnb listings in LA**.

How has the Airbnb Market in LA Changed after COVID?



Data Pre-Processing

- The graphs only include the **top 3 clusters** discussed in the previous slide, focusing on the most significant trends in our analysis.
- The data was normalized using z-score normalization to facilitate trend analysis, and the results of our analysis are presented on the left.

Pre-COVID (Jan 2019 – Feb 2020)

- Stable growth in listings and reviews as Airbnb demand increased.
- Prices remained relatively steady, reflecting normal market conditions.

Lockdowns (Mar 2020 – June 2020)

- Sharp drop in total listings & reviews due to travel restrictions.
- Prices fluctuated as some hosts lowered rates to attract local travelers.
- Many hosts paused or exited the market due to uncertainty.

Uncertainty Phase (July 2020 – Dec 2020)

- Gradual recovery in listings as travel restrictions eased.
- Reviews increased, signaling a return of demand, but still below pre-pandemic levels.
- Prices remained volatile, influenced by changing regulations and travel behaviors.

The COVID-19 pandemic caused significant disruptions in the Airbnb market, with listings and demand dropping sharply during lockdowns and gradually recovering in the uncertainty phase. As travel behaviors shifted, hosts had to adapt pricing strategies and reassess market conditions.

How Government Regulations and COVID-19 Shaped Airbnb Trends

The same data pre-processing procedures from the previous slide were applied to ensure consistency in trend analysis.

Pre-COVID (2019 – Feb 2020)

- Los Angeles (Nov 2019) required Airbnb listings to display a city-issued license for short terms rentals which may have caused availability to increase
- Total listings stayed stable, but stricter regulations caused some listings to drop off the market.
- Entire homes were more impacted by regulations while private rooms were less affected

Lockdowns (Mar – Jun 2020)

- COVID-19 travel bans and lockdowns led to mass cancellations and fewer active listings. as a result availability spiked as travel demand collapsed.
- Entire Homes and Private rooms experienced sharp increase in availability as bookings started dropping.

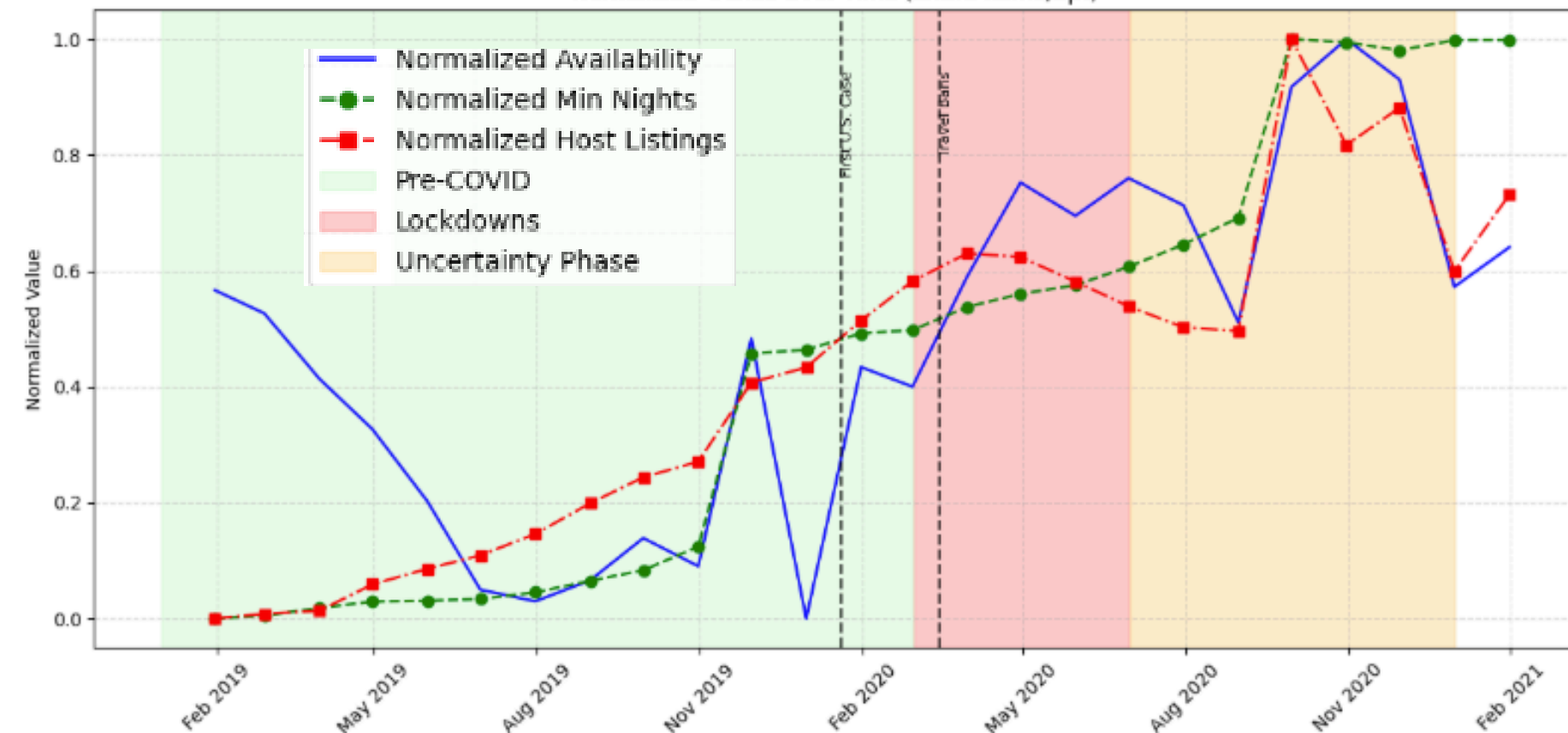
Uncertainty Phase (Jul – Dec 2020)

- Availability fluctuated as travel resumed but remained inconsistent.
- Minimum nights experienced a sharp uptake in September 2020 and stayed consistent afterward, possibly a result of the city regulations

Conclusion

- Regulations and COVID-19 permanently changed Airbnb's landscape.
- Hosts shifted toward long-term stays, adapting to new traveler behaviors.
- Private rooms saw higher levels of volatility in average price and host listings drop post Covid
- The future of Airbnb depends on how cities regulate short-term rentals and how hosts adapt.

Normalized Trends Over Time (Entire home/apt)



Normalized Trends Over Time (Private room)

